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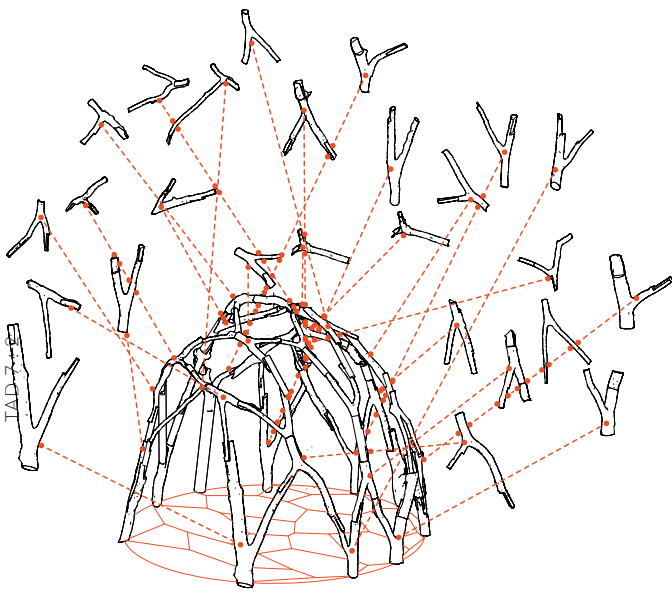
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Wild Wood Gridshells: Mixed-Reality Construction of Nonstandard Wood



Irregular wood is often downcycled despite having significant embedded strength. Reintegrating this wood into structural assemblies can improve material efficiency in the built environment. This work implemented material logic in a design-to-fabrication workflow for building structures using bifurcated tree branches to leverage this potential (Figure 1). This process is demonstrated through the design and construction of a prototype. A user-oriented computational interface is proposed that manages irregular geometries, matching and optimization algorithms, and structural simulation for design iteration. The demonstrated workflow, which concludes with augmented reality (AR) assisted fabrication, facilitates designing with varying materials, enabling upcycling a wide range of nonstandard building elements. At scale, this methodology can significantly reduce the environmental impact of construction.

◁ Figure 1. Wild wood assembled in a gridshell. (Credit: Authors for all figures)

Keywords: Material Upcycling, Augmented Reality and Fabrication, Material Assembly and Manufacturing, Structural Design

Introduction

Motivation

The construction sector plays a significant role in global greenhouse gas emissions, accounting for 37% of worldwide energy and process-related carbon dioxide (CO₂) emissions in 2021, with around 9% coming from the production of building materials.¹ The IPCC report points out that the carbon footprint of buildings must be drastically reduced.² Improving material efficiency and implementing circularity are two opportunities to reduce embodied carbon.³ Further, the reuse of materials for structural applications has the potential to mitigate the environmental impact of construction significantly.⁴

Current wood-building practices rely almost exclusively on highly standardized products such as dimensional lumber. This paradigm limits the industry's ability to use available materials such as timber processing by-products or upcycled elements. Wood pieces and cutoffs characterized by small diameters, knots, kinks, bifurcations, and bark, cannot be standardized by the lumber industry and are downcycled or sent to the landfill as waste wood. In 2018, the US Environmental Protection Agency estimated that the timber industry yields 57% usable lumber and 43% waste material in a harvest.⁵ Recovery and reuse of waste wood is a crucial strategy of the EPA's Sustainable Materials Management efforts.

Building with recovered harvest waste wood, with diameters ranging from two to six inches, has an environmental carbon storage potential, sealing that wood's carbon in architectural form and preventing it from being released into the atmosphere. From an economic standpoint, this maximizes the utility of felled trees. Though many mills use all waste material, their outputs in those cases have a lower value return than those used in structural applications. While there is an ecological benefit to wood decomposing on forest floors, this work argues that we must balance ecological benefits with climate determinants and carbon release. There will always be small-scale waste wood (less than 1"–2" diameters) that can contribute to ecological forestry cycles. An alternative ecological approach to building with small-diameter tree forks encourages selective trimming rather than complete tree removal. Trimming allows trees to continue to live, grow, and sequester carbon. This alternative forestry approach would allow harvesting in shorter cycles, support tree growth, avoid detriment to forests, and encourage the absorption of atmospheric carbon. Beyond forest harvesting, the valorization and upcycling of trimmings from urban trees, often deemed waste wood, could have significant environmental benefits.⁶ This wild wood has valuable material properties that could be leveraged in structural assemblies.⁷ Bifurcated branches, for example, are naturally optimized during tree growth, and consequently their strength rivals that of manufactured wooden joints.⁸

Furthermore, round timber can be up to five times stronger than the largest piece of dimensional lumber the same branch could yield, meaning small diameter, round branches are valuable structural members.⁹

This research aims to leverage the embedded structural strength of tree forks, defined as bifurcated tree branches, with a computational design and fabrication workflow. The project develops methods for structural design that take advantage of nonstandard elements and fabrication strategies that accommodate complex assemblies. Although this project focuses on upcycling wild wood, the design and fabrication methodologies presented can be adapted to develop efficient structures using most nonstandard elements, effectively enabling new forms of material circularity across the industry.

Related Work

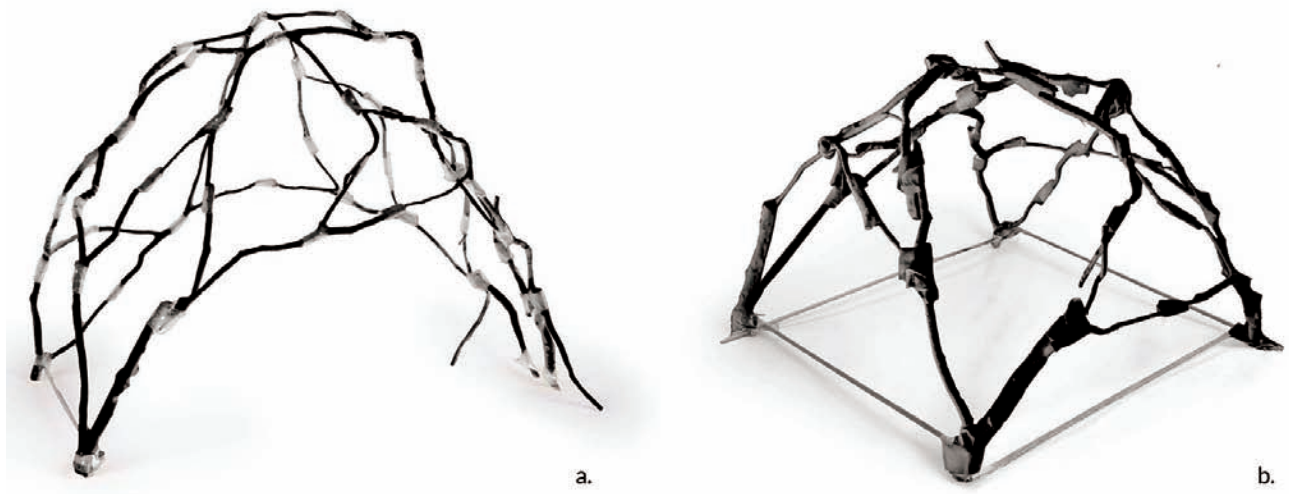
State of the Art

In recent years, there has been an abiding interest and a growing body of work on designing structures with natural wood, small-diameter wood, tree forks, and irregular wood. An overview is provided in "Whole Timber Construction: A State-of-the-Art Review."¹⁰ AA Hooke Park is renowned for innovative research combining new forestry practices with digital architectural fabrication. The "Tree Fork Truss" demonstrates the potential of custom fabrication workflows to create structures with tree forks.¹¹ Within the respective computational methods, however, there is little emphasis on design flexibility or opportunities for design iteration. The "LIMB" project at the University of Michigan addresses design integration and intention, but the tradeoff for more design flexibility is more digital fabrication and intensive machining.¹²

There are several strategies for designing irregular tree forks. One strategy consists of using a target design to which elements are mapped. This method is employed by the previously mentioned Tree Fork Truss and LIMB project and in the research presented in this paper. Another strategy is to allow the elements to guide the design using an aggregation logic. This approach is explored by the "Conceptual Joining" research at the University of Applied Arts Vienna¹³ and, more recently, in Mikio Koshihara's lab at the University of Tokyo.¹⁴ This method gives the design more agency in accommodating the unique geometries of each fork, but control of the overall form is limited.

The Digital Structures research group at MIT has studied strategies to harness the structural strength of tree forks in structural assemblies.¹⁵ This research introduced a real-time computational matching method using the Hungarian Matching algorithm and the optimization of the structural network to improve assignments between tree forks and the target nodes.¹⁶ The fabrication method minimized joint manufacturing time by shaping with a bandsaw rather than milling. Although this digital fabrication process is less intensive, it requires a robotic fabrication setup.

Beyond robotic fabrication, recent work has leveraged the versatility of mixed reality (MR) fabrication strategies. In "Making in Mixed Reality," the founders of Fologram introduce holographic registrations for precise fabrication guides, enabling new



△ Figure 2. (a) Aggregative method; (b) topology matching method.

protocols that reduce safety requirements, hardware expertise, and equipment costs compared to robotic or CNC processes.¹⁷ The method is fully scaled in the Steampunk Pavilion at the Tallinn Biennale, designed by Soomen Hahm and Igor Pantic.¹⁸ Timber De-Standardized at Cornell University brings attention to the specific potential of MR fabrication for found and nonstandard wood assemblies. Their work proposes an MR user interface to increase design agency within the constraints of a catalog of distinctive elements.¹⁹ Researchers of the ETH spinoff INCON have developed other MR solutions enhanced with computer vision and machine learning algorithms to improve AR overlays, enabling increased precision in fabrication.²⁰ Teams at Gramazio Kohler Research have used it to assemble the “Touch Wood” acoustic wall.²¹ The versatility in working with distinctive elements and the potential to achieve precision in fabrication that these precedents in MR fabrication research suggest indicates that these tools are well suited for constructing structural assemblies using irregular tree forks.

Research Gap

The references outline a clear potential for reusing nonmarketable irregular wood elements like tree forks in new structural assemblies and provide tools and strategies to work towards that goal. However, the research presented in this paper seeks ways to scale up from one-off research projects and unique digital fabrication setups, identifying two missing conditions in working towards that goal:

(1) An intuitive design methodology: The complexity of designing with irregular elements needs to be simplified into a systematic approach supported by comprehensive, accessible, and versatile methods leveraging a combination of human intuition and automated computation.

(2) An accessible and replicable fabrication strategy: Beyond high-end fabrication infrastructures and leveraging the development of MR solutions for construction, the method needs to provide flexibility in fabrication and tooling processes without requiring extensive capital investment at any step.

New Contributions

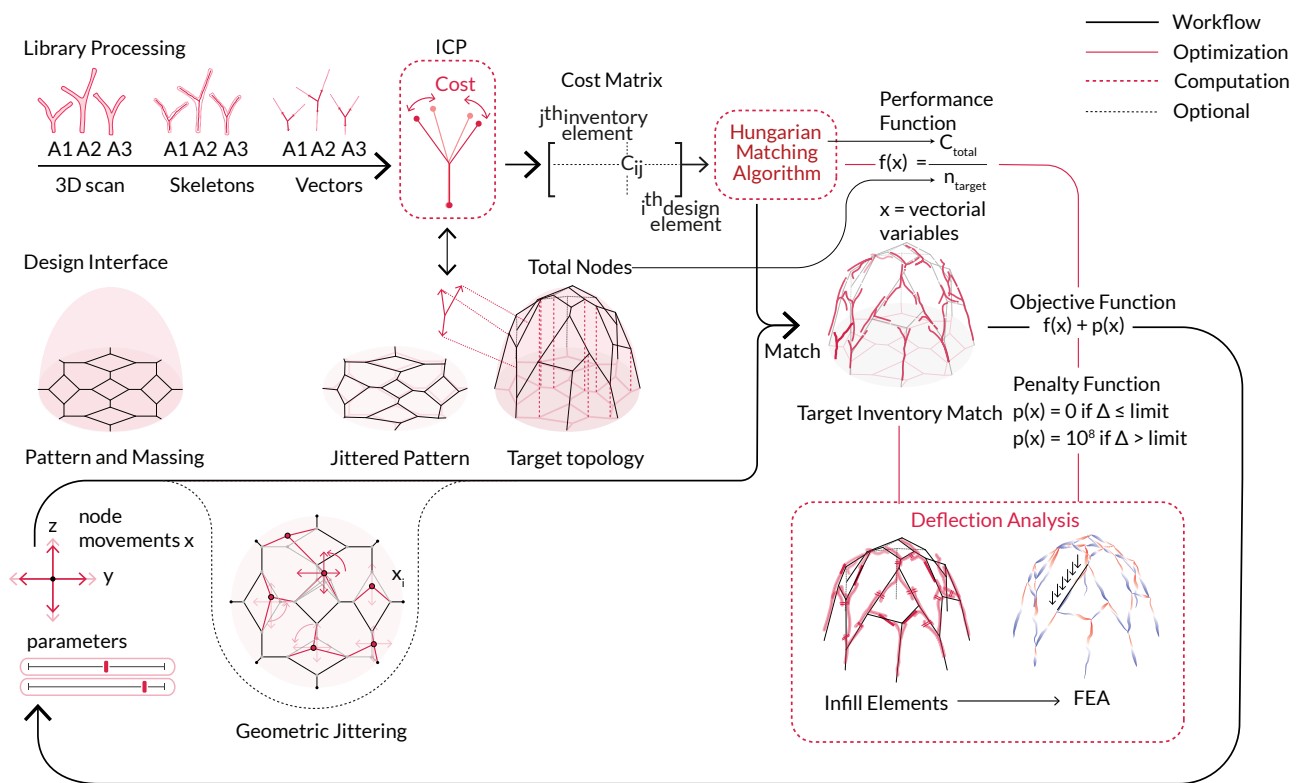
1.3.1. Design Methodology that Balances Design Intuition and Computational Problem Solving

This paper presents a design methodology for wild wood structures that accommodates specific design intents with automated computation through a flexible, interactive, dynamic workflow and interface. To understand the challenges of designing a structure with tree forks whose bifurcation angles vary widely, scaled physical models were made with a 3D-printed inventory of the tree forks. Manual assemblies were built by different team members using scaled forks and tape for joints. From this modeling, two different intuitive design strategies emerged:

(1) The first is a freeform incremental aggregation of forks from a chosen starting point, intuitively accounting for past angles and anticipated future form. In this approach, illustrated in Figure 2, it is difficult to predict the overall massing and topology. This method is bottom-up, driven by the forks.

(2) The second approach is top-down, defined by a topology. A planned pattern organizes forks, as Figure 2 shows. This process proceeds iteratively from one or several starting points, like a greedy search algorithm. The end massing and topology are predictable, but completing the design is difficult because angular deviation from the target pattern accumulates as the assembly progresses.

A successful design workflow using tree forks (1) leverages the individual geometries of each element and (2) gives designers agency over the design of the massing and the topology of the structure. This research aimed to develop a process emphasizing collaboration between human intuition and computational tools, reconciling these objectives and enabling designers to



△ Figure 3. General overview of the computational design workflow.

work with irregular materials. The computational design methodology proposed emphasizes the design interface, allowing designers to browse and manipulate different topologies with real-time visualization of the resulting tree fork structure. In short, the first contribution of this paper is a workflow that simplifies designing with tree forks and allows a wide variety of structures to be conceived, as demonstrated through several wild wood gridshell designs.

Accessible Fabrication Strategy

Designing a structure with tree forks poses a challenge regarding physical assembly with nonstandard elements. In most of the previously mentioned work (see above, 1.2), robotic and digital fabrication (five-axis CNCs) is used to manufacture unique assembly connections accurately. This approach and equipment are limited to dedicated research laboratories or high-end fabrication spaces. Though the precision is advantageous, the difficulty of accessing such equipment is a constraint that hinders real-world replication. This research posits that accuracy with intensive digital fabrication could yield more versatile and accessible fabrication for a more approachable design-to-build workflow.

This paper presents a generalized tooling strategy that does not require robotics and leverages mixed reality to enable the fabrication and assembly of irregular wooden elements into complex structures using only accessible woodworking tools.

The fabrication workflow is demonstrated through a tree fork gridshell, assembled by three researchers with manual saws, an electric bandsaw, drills, and basic woodworking tools.

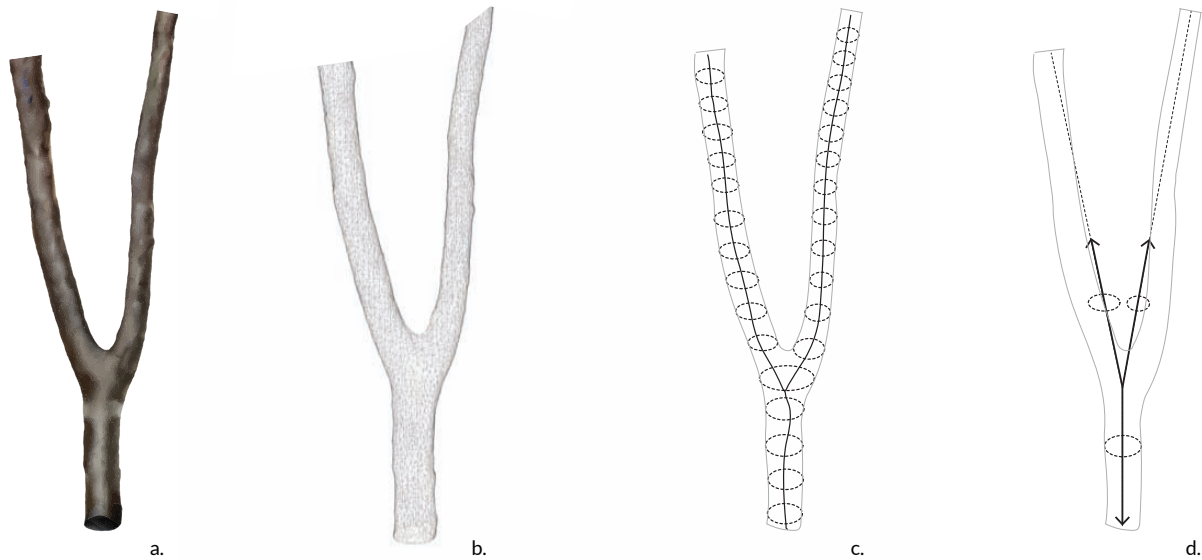
Methodology

Conceptual Overview

The proposed workflow initiation is twofold:

1. Scanning salvaged tree forks into a digital inventory where data processing can be performed.
2. Digitally iterating through designs to select a target topology onto which the tree-fork inventory can be matched.

Overall angle differences between tree-fork chords and assigned nodes in the target network are then computed using the ICP (Iterative Closest Point) Algorithm. Next, the Hungarian Matching Algorithm uses the cost matrix produced by the ICP to find the optimal matching solution characterized by a total cost. A lower total cost equals a lesser angle discrepancy between the tree forks and the target network. According to the determined match, the tree fork meshes are mapped onto the digital target network. Where gaps exist between assigned tree forks, unprocessed linear wood segments are used as an infill to connect neighboring tree forks. The digital assembly, composed of tree forks and linear segments, is fed



△ Figure 4. (a) Fork from the inventory; (b) 3D-scanned mesh; (c) extracted skeleton and diameters; (d) simplified vectorial representation.

into a structural analysis model to evaluate structural viability. Finally, an optimization loop is created using total matching cost as an objective and parametrization of the target network topology as parameters. The optimization loop runs, jittering the target network nodes, evaluating structural performance, and refining the match quality (Figure 3). The digital model of the final allocation is then used to support MR fabrication and assembly of the physical structure.

Digital Inventory

Sourcing

Tree forks and small diameter linear segments (2"–6", 5–15 cm) are easily sourced from storm debris and the annual thinning of urban trees as part of municipal management. The unmarketability of this wood means it is waste, and those in charge of disposal are keen to give it away to anyone willing to pick it up. It is important to note that sourcing useful wooden elements requires quality control. One must ensure the wood has not started to rot, which would make it unusable in a structural assembly.

Inventory Processing

The first step of the computational design workflow is to 3D scan collected tree forks into a digital inventory. The digital meshes are then imported into Rhino/Grasshopper (GH), where key information is characterized. First, tree fork skeletons (mesh centerlines) are extracted using the mesh contraction method provided by the GH plugin Cockroach.²² Skeleton nodes and endpoints generate vectors of each tree fork's three chords. Next, skeletons and mesh analysis are used to find average chord diameters for each tree fork (Figure 4). The inputs utilized throughout the computational process are the vector angles and chord diameters.

Design Interface

Design Interface

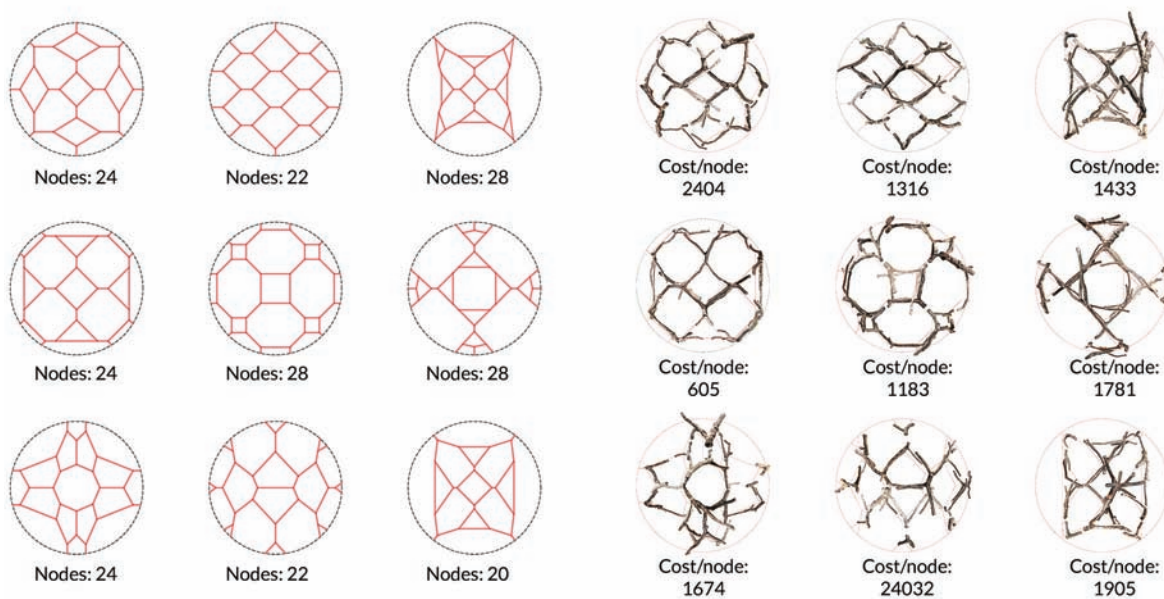
The proposed Rhino/GH design workflow allows the designer to define the overall massing through a 2D pattern topology projected onto a surface. Tools like the GH Plugin Parakeet allow for simple pattern generation that the user in Rhino can easily manipulate.²³ These tools enable iterating through different design options with real-time visualization of the tree fork matching on the target network (Figure 5). The examples below illustrate various patterns and fork matching in a plan for gridshells. The design workflow is not limited to gridshells, though some topological constraints are linked to the three-chord nodes, which gridshells can easily accommodate.

Real-time Inventory Matching

Like the tree forks in the inventory, each node of the target pattern is characterized by the three vectors extending from each node. The three vectors characterizing each tree fork and the three vectors from each target node are cross-analyzed and compared using the ICP Algorithm. Every possible match between a tree fork and a target node is assigned a score, or "cost," that characterizes the angle misalignments. The higher the ICP cost, the more variation between the two vectorial sets. If there are n tree forks in the inventory and m target nodes in the design, the ICP costs are compiled into an $n \times m$ matrix which characterizes all possible tree fork to target node combinations. The computed cost matrix is then input into the Hungarian Matching Algorithm, which determines the optimal assignments, minimizing the total cost.²⁴

Infill Elements Generation

The matching process results in a configuration of tree fork meshes mapped on the target network. In some cases, tree forks



△ Figure 5. (a) Example of patterns; (b) real time fork allocation with matching scores.

overlap and can be joined directly. In other cases, a gap between two neighboring forks means an unprocessed linear wood segment is introduced as an infill to complete the structure (Figure 6). The developed workflow generates the linear segments within the digital model and approximates the ideal infill diameter based on the neighboring chord diameters.

Optimization of the Network

Total Cost

The total cost, output by the Hungarian Matching Algorithm, characterizes the match quality between inventory elements and target network nodes. A high-cost design is characterized by more misalignments between tree forks and the target network, and minimizing this cost equates to improving the match and achieving an assembly visually reminiscent of the designed topology. An optimization loop has been embedded in the computational workflow where minimizing matching cost is the objective, and the target nodes' vector translations (jittering) are the parameters to achieve this.

2.4.2 Network Jittering Parameterization

Displacement of the nodes, or vertices, in the 2D pattern adjusts pattern angles and thus the three-chord vector arrangement of target nodes, while maintaining the overall topology. The vertices are asynchronously moved in the **x** and **y** direction, parameterizing the design and allowing a minimal cost to be found. Displacement of the vertices in the jittering process is achieved with two parameters for 2D translations. Therefore, if there are **n** nodes in the network, there are **2n** parameters. With an average network of 24 nodes, this equates to a computationally heavy optimization with 48 parameters. To optimize more efficiently, the dimensionality of this problem is reduced by creating two proxy surfaces. Each

proxy surface is defined by five control points (four corner points and one center point). This optimization reduces the number of input parameters from 48 to 10. The two surfaces are then sampled to produce **z**-coordinates associated with each **n** node. Each surface assigns translation in one direction. The **z**-value distributions of the sampled points for each surface are remapped to the pattern nodes informing displacement (Figure 7).

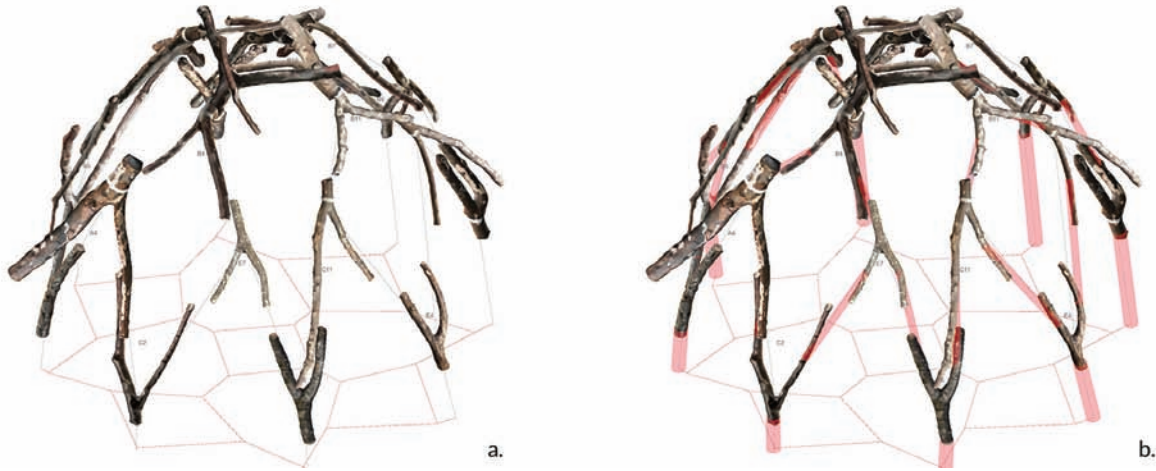
Structural Analysis

In parallel to the matching process, structural analysis is performed on the output design iterations with information extracted from the tree fork mesh geometries (Figure 8). Tree fork skeletons are divided into one-inch segments, and for each interval, the diameter of the tree fork's mesh is sampled. These diameters inform the cross-section for structural analysis with Karamba for Rhino/GH.²⁵ This allows the structural viability of the design to be evaluated and provides additional information on the feasibility of the design being explored. Theory and experiments (Desai, *Thesis*) support assigning bending moment resistance to the tree forks and scarf joints. The structures, thus, do not need to be purely funicular, as they can support nonaxial forces. Structural analysis using Karamba3D models rigidity through fasteners representing dowels in the scarf joints. The deviation computed in the structural analysis defines a "penalty function" compounded in the optimization's objective function. As the optimizer iterates, if the defined deviation threshold is exceeded, the penalty is applied, and the algorithm moves away from this solution.

Joint Strategy and Manufacture

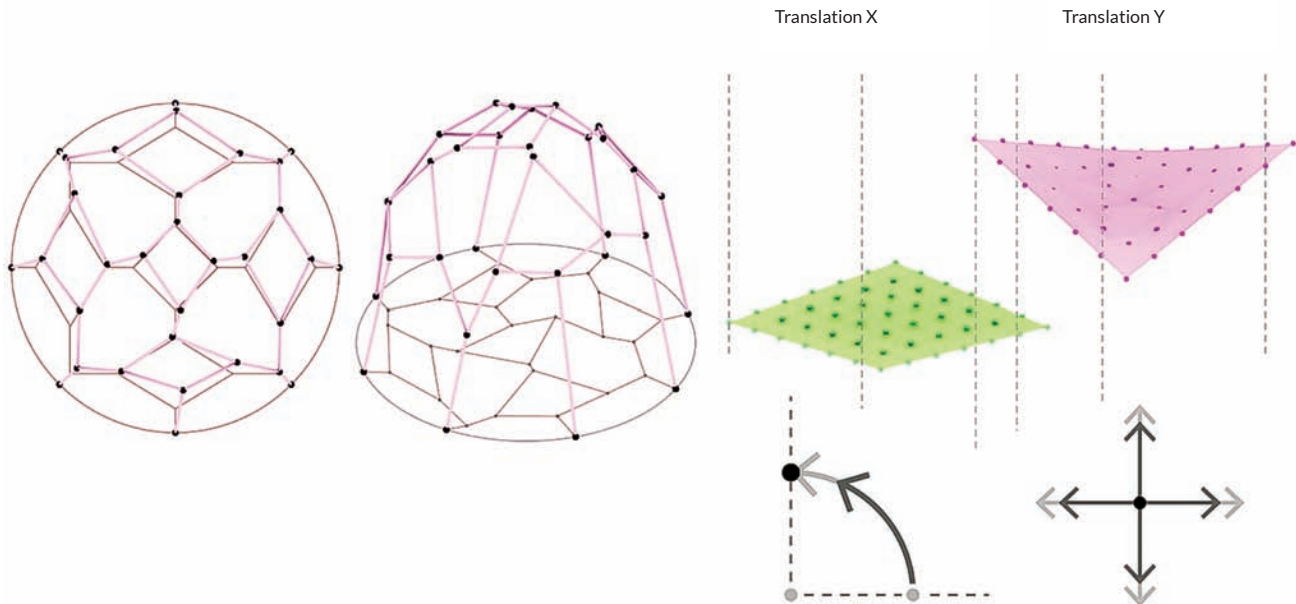
2.6.1 Joint Typology

After testing the strength and fabrication of several end-to-end joints, a simple scarf joint with dowel connections was selected to



△ Figure 6. (a) 3D model of matched forks; (b) generation of infill elements.

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△ Figure 7. Network jittering with displacement parameters generated by the manipulation of proxy surfaces.

assemble the tree-fork structure. Scarf joints can be easily manufactured with just three saw cuts and dowels inserted at several angles opposing the joint's cut line, locking the joint in place and providing a bending moment connection.

Joint Modeling

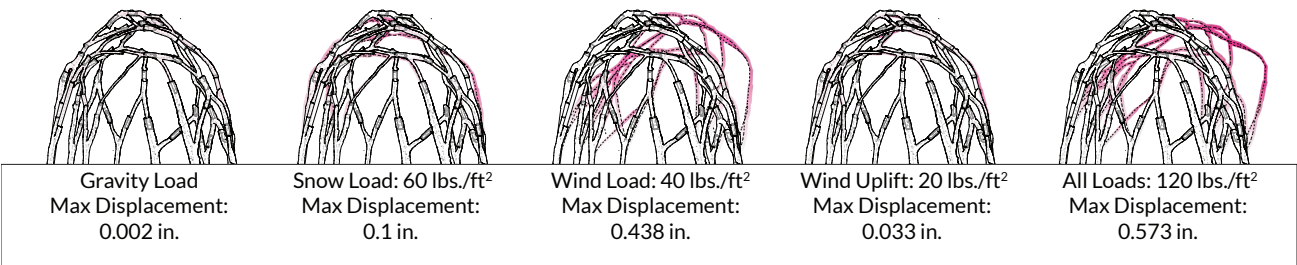
Tree-fork skeletons and meshes are referenced in the digital model to design the joints. A line is drawn between two forks from an 8 cm inset from skeleton endpoints (Figure 9). A rectangular bounding box is then generated that encloses this line, and the branch ends. This bounding box is cut in half, each half used to trim one of the two tree-fork meshes. This results in a digital scarf joint with trimmed meshes indicating how the real tree forks must be cut.

Joint Manufacture

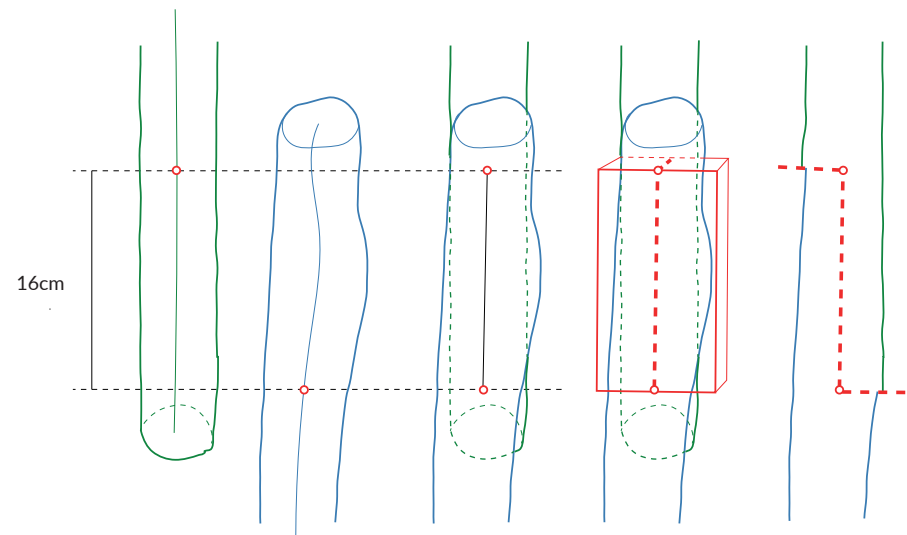
A mixed reality workflow was developed to cut the scarf joints of irregular tree forks without using a heavy fabrication setup. This process involves projecting the AR digital model of trimmed meshes onto individual tree forks. Looking through a smartphone, one can easily see the location of the scarf joint and trace the joint's cut lines directly on the tree fork with a pen (Figure 10). Once the cuts are traced, they can be cut with a band or manual saw. The cutting tool depends on the size and geometry of the fork relative to the cut.

Joint Tolerance Strategy

Compared to an open-loop digital fabrication workflow where all tree forks are precisely prefabricated, the tolerance of this



△ Figure 8. Structural analysis of the matched fork structure under various load cases.



△ Figure 9. Scarf joint modeling protocol using the fork meshes and centerlines.



△ Figure 10. Mixed reality joint tracing.



△ Figure 11. Typical scarf joint with plastic buffer.

closed-loop MR fabrication process is relatively high. With the AR joint cutting processes, some scarf joints don't perfectly align on-site, making it difficult to secure the joint for drilling and inserting dowels. A buffer material is introduced as a "soft joint," using biodegradable plastic that is melted in hot water and sandwiched between joint cuts to address these tolerance concerns (Figure 11). Once it cools, the buffer temporarily secures the tree forks, allowing holes to be drilled and dowels inserted. Further, the biodegradable plastic acts as a seal, preventing water penetration into the exposed end grain. The assemblies proposed in Gil Sunshine's Medium Resolution inspired the use of bioplastic as a joint interface of two nonstandard structural elements.²⁶

Assembly of the Structure

Assembly Process

The irregular nature of the wild wood assembly poses a challenge in accurately positioning and assembling tree forks on-site. In an analog construction setup, one approach to referencing tree fork positions during assembly would be jiggging that accurately locates points in the tridimensional assembly space and temporarily supports the in-progress structure. An MR construction space using Fologram avoided a complicated construction setup and enabled a more agile assembly. This approach consists of a fixed QR code on-site, which anchors a digital projection of the target assembly. The construction assembly requires at least three people. The first carries and holds tree forks in place. The

second communicates the accuracy of the positioning and alignment with the digital projection on a portable device (smartphone or tablet). Once the tree fork is in position, the third applies the bioplastic buffer and compresses the joint while the buffer cools (about five minutes). Slight positioning corrections can be made at this time. Once the buffer is cooled and the tree fork temporarily secured, holes can be drilled and dowels inserted across the joint, with slight angle differences, providing structural integrity and locking the joint in place.

Adaptive Design and Assembly

There is an order of assembly from the outer, lower tree forks to the higher, apex tree forks. Therefore, if an element were found to be defective at any point throughout the assembly, the fabrication could be paused. In this scenario, there are two courses of action. The first would be to refer to matching allocations and determine which tree fork was designated as 'second-best' for the node assigned a defective element. If this alternative piece is not satisfactory from a design perspective, a second approach would be to identify which tree forks have already been assembled and re-iterate through the computational workflow matching only the remaining unassembled nodes with the remaining unused tree forks. Already assembled and placed tree forks and target nodes would be fixed in this scenario; only the parameters of the remaining unassembled nodes would be iterated. Thanks to the flexibility and high tolerance of the fabrication strategy, the first round of construction would blend seamlessly with the following rounds of matching and assembly.

Results and Applications

Prototype Structure

Prototype for a Solar Kiln

The design-to-fabrication workflow presented in this paper is demonstrated through a wild wood gridshell prototype that will serve as a solar kiln for an MIT campus woodshop to dry timber. This function defined the structure's massing, a simple half-ovoid with a height of eight ft. and diameter of 10 ft. The wild wood gridshell is clad with reused acrylic panels, allowing the sun to enter, heat the interior, and dry green wood.

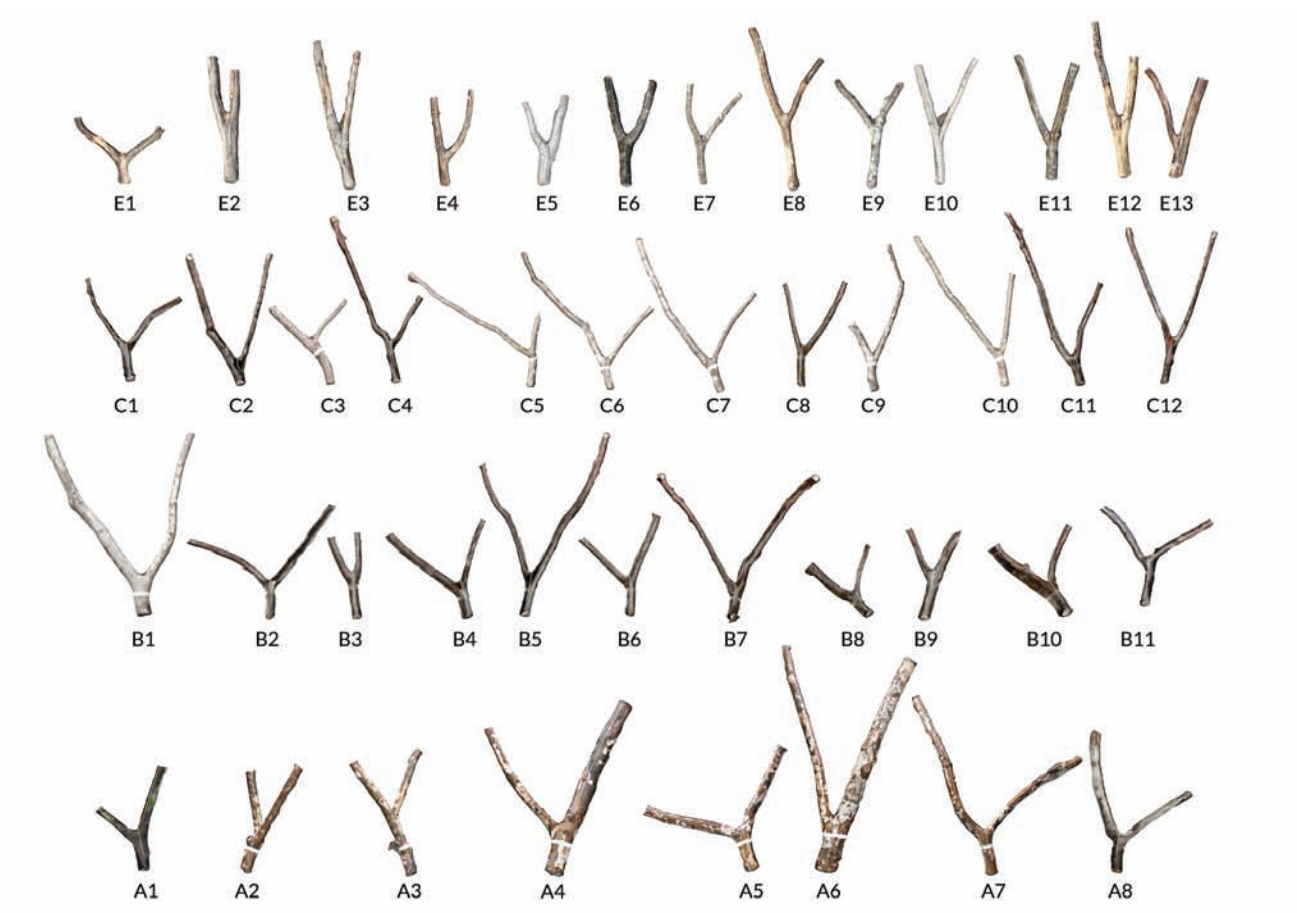
Sourcing

With the help of the MIT Office of Sustainability and Campus Construction, wooden offcuts were collected from trees removed for the construction of a new campus building and from maintenance trimming (Figure 12).

After sorting out forks that were too irregular or whose diameters were too small ($<2"$), the research inventory comprised 44 tree forks that were then 3D scanned using photogrammetry with smartphone camera pictures with Metashape and characterized to constitute a digital inventory²⁷ (Figure 13). A larger collection of unprocessed linear segments was also collected, with diameters ranging from 2" to 6". These linear elements were more plentiful than tree forks in the collected thinning, and thus easier to source.



△ Figure 12. Tree branch pruning in MIT's Killian Court.



△ Figure 13. 3D-scanned inventory of forks (44 elements).

Design with a Constrained Inventory

The pattern for the final design was chosen with consideration for the external cladding system. To reduce the spans of the cladding substructure, a pattern that reduced void spans in the gridshell—in other words, a denser network—was preferable. However, the inventory size was a limitation to increasing the density of nodes in the topology. The chosen design thus balances these two constraints with several nodes slightly less than the inventory size, allowing some flexibility in the allocation while maximizing the inventory usage and network density. The selected design iteration has a topology with 38 nodes, and the inventory only contained 44 elements; this was a constrained inventory design problem with 86% of the elements used.

The matching was done in two rounds to improve the distribution of diameters across the structure. The first used the outer portion of the 2D pattern (or the lower half of the target network) and matched larger diameter tree forks from the inventory. The second used the central portion of the 2D pattern (or the upper half of the target network) and matched the remaining tree forks from the inventory. The pattern was optimized for each round to improve the match using “Radical” from the DSE Suite.²⁸ The tree forks were then trimmed in the digital model to avoid overlaps, and infill elements were generated to bridge the gaps between neighboring tree forks (Figure 14). Infill elements were sourced from collected unprocessed linear segments. Finally, digital scarf joints were modeled.

Assembly

The fabrication of the pavilion took 15 full days of work, with a total of 300 hours of work (Table 1). An approximate breakdown of hours can be found below. Debarking tree forks was time-consuming and could be improved with more tailored hand tools, waiting for the right seasonal conditions, or doing so quickly after trimming. Tracing cut lines via AR was the second most time-consuming step and could be improved with a more efficient AR setup. Assembling the tree forks on site (Figure 15) was quick in comparison. Preparing the forks (debarking, tracing, cutting, finishing) took over three times more than assembly.

Results

The structure was completed based on the digital model (Figure 16); the deviation between the final assembly and digital model, assessed in the MR environment, ranges from one to four inches. The greatest deviation is at the top of the structure where the tolerance adds up; nonetheless, this was not an issue as the last element was assembled in place correctly. The built structure is resistant and stiff enough to carry a person’s load from the top without any visible deflection or ruptures (Figure 17). Attempts were made to scan and compare the built gridshell with the digital model; however, technical issues were met when scanning the structure in a challenging environment.

Reflection on the Process

The work in this study aimed to establish an accessible and flexible workflow to construct a gridshell with irregular tree forks. Our process successfully used a limited tool set (smartphone

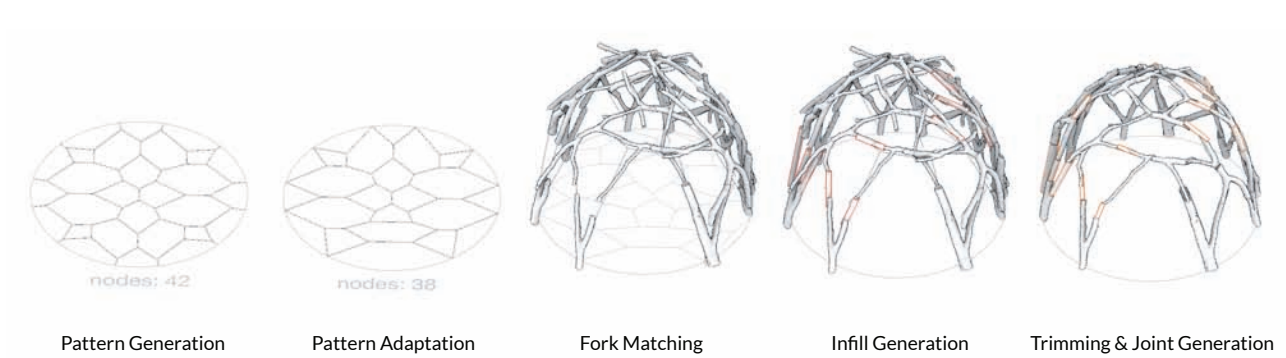
Table 1. Amount of required people-hours for each task.

Task	Time (person-hour)
Debarking	85 h
Tracing in AR	65 h
Cutting	25 h
Finishing	15 h
Preparing buffer soles	20 h
Assembling on-site in AR	55 h
Other	35 h
Total	300 h

and saw) and an inventory of elements to build a structure with design intent maintained. The design workflow proved flexible and efficient enough to compose the desired structure with a highly constrained inventory of irregular parts. The fabrication strategy allowed for tolerance discrepancies, easing construction with the irregular material inventory. Assembly was not challenging, and the fabrication strategy proved effective, as no major element modifications were needed throughout construction. Woodworkers—two novices and one experienced—constructed the gridshell within 300 person-hours. Arguably, the labor involved in repetitive tasks could be reduced with robotics. However, the use of robotic fabrication for joinery seen in related research demands extensive and complex setup time and calibration due to the irregular nature of each piece. The advantage of the method used here was not only the elimination of setup time but that multiple processes could coincide. Fabricators can work in parallel, each responsible for tolerance management compared to the linearity imposed by one or a few robotic agents. Although the final assembly could not successfully be scanned to measure the geometric deviation in all points, this alternative fabrication methodology certainly has a higher tolerance than fully automated workflows. The tolerance increased during the tracing and cutting of the fork’s joints and when aligning the forks with their AR final location projections during assembly. However, the closed dome structure could still be completed, which is a much more exacting system than an open-ended wall assembly where tolerance can add up in one or several directions with fewer consequences. Thus, the higher tolerance in the proposed workflow is successfully managed, making it viable for construction.

Design Possibilities

The proposed workflow allows a large range of structures and topology patterns to be explored. Figure 18 demonstrates a variety of potential designs that can be assembled with tree forks using the same methods and inventory. Most tree forks, as do those used in this research, have three chords branching from a central node. Many lattice-based structures, such as trusses, however, have many elements converging at each node, meaning they can’t be produced with a simple end-to-end



△ Figure 14. Main steps of the digital design process.



△ Figure 15. Pictures of the MR assembly process.

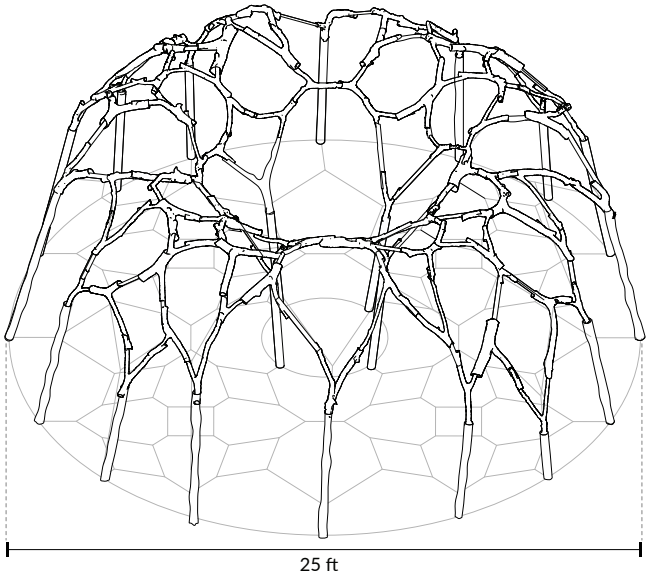


△ Figure 16. Final assembly of forks into a gridshell.

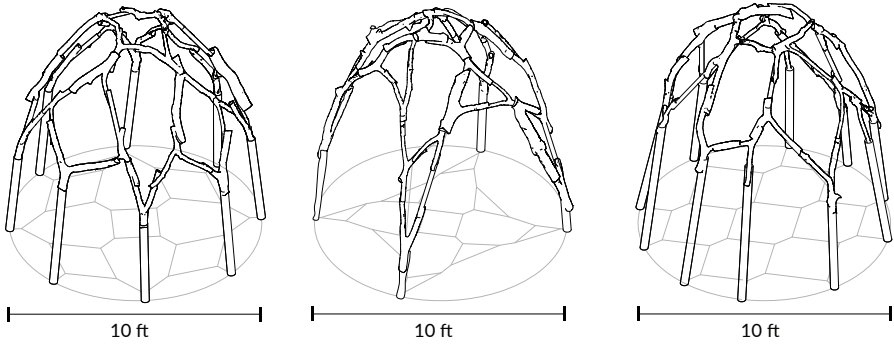


△ Figure 17. Applying loads on the structure.

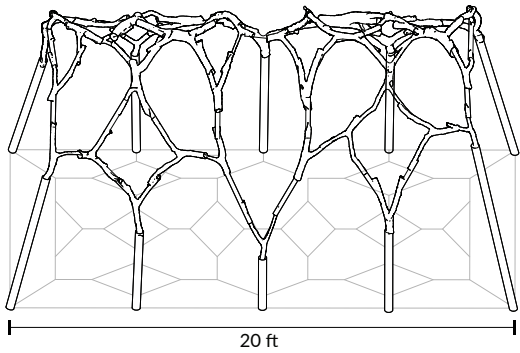
Ring Configuration



Circular Configuration



Rectangular/Linear Configuration



◁ Figure 18. Further potential fork assembly designs.

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assembly of tree forks. This topological limitation oriented the target design towards gridshells that, in principle, are challenging to construct due to the multi-element nodal connection points. Connecting two elements is significantly more accessible than connecting three or more elements at precise angles. The naturally occurring three-chord nodes of the tree fork thus facilitate the construction assembly because connections can be made one-to-one. Despite multiplying the number of fabricated joints by 1.5, each joint is more easily fabricated and assembled. Gridshells are well suited for this workflow, but in future iterations, further lattices like spatial trusses and arborescent structures would also be good candidates to investigate.

Discussion/Future Work

Variable Trim

Wild wood comes with very diverse and irregular geometries. The tree forks sourced for this prototype had several kinks on each chord where the angle of the branch would change. This challenge was simplified by trimming branches at their first kinks. The possibility of trimming the tree forks at different kinks as a parameter was explored, as this would enable the inventory to expand, each fork yielding several vectorial representations due to trimming at variable lengths. Considering the constrained inventory, the greater variety of angles in this expanded inventory could be precious. Still, it presents the challenge of managing tree fork versions and assignments while complicating fabrication and assembly. Resolving these computation and fabrication questions, however, could enable the unique geometries of tree forks to be further leveraged and increase the variety of structural recombination they can compose.

Multi-objective Optimization

In the presented research, the single objective optimization of the network is the Matching Cost derived from the Hungarian Matching Algorithm. Other variables that could prove effective are diameter distribution, structural capacity (derived from the in-loop structural analysis), and length of infill wood (in cases where we want to use only forks to complete the structure). These variables were integrated into a Multi-Objective Optimization (MOO) using the DSE suite for Rhino/GH.²⁹ The results showed that with such a constrained inventory, the tradeoff between different objectives was not favorable for the quality of angular matching and yielded challenging structures to build. Future work on larger inventories would be an excellent opportunity to introduce MOO and weigh several objectives, including diameter distribution and structural capacity.

Fabrication

Time and labor are important considerations for the construction of similar structures. For example, tree fork processing is much more time-intensive than assembly. One solution would be to have a larger construction team, where one group processes forks and a smaller group assembles simultaneously, to reduce construction time. Another challenge of this construction method is digital-to-physical discrepancies. The accuracy of the AR fabrication process could be improved by switching

from smartphones to larger tablets or even using AR headsets. In Fologram's new version, feedback is provided by the HoloLens using the device's depth camera, informing users on the quality of alignment between the digital and physical models through display features such as colormaps.³⁰ This feedback could ease the assembly process and improve overall fabrication precision in larger-scale works.

Scaling-up

The prototype uses relatively small tree forks (diameter between two and six inches). Similar structures could be built at much larger scales, but the increased scale would require heavier equipment to collect, machine, and assemble the elements. A similar AR workflow applied to the machining and assembly of nonstandard wood elements at a larger scale has been demonstrated in projects such as the Negami-no-Takumi workshop, during which bent tree trunk columns were fabricated using chainsaws and AR headsets.³¹ Computer-vision-enhanced AR solutions like those mentioned in section 1.2 could also enable more precision to be achieved and thus improve the workflow's applicability to larger-scale applications with more refined detailing.

Conclusion

Summary of Contributions

This paper proposes two critical contributions to systematize the methodology from design intent to fabrication, building upon the existing body of research on the computational fabrication of wild wood structures and design strategies for irregular elements upcycling.

A design methodology for wild wood structures that accommodates specific design intents with automated computation through a flexible, interactive, and dynamic workflow and interface. This methodology includes a digital approach to characterizing and managing inventories of irregular elements, a computational method for generating and manipulating grid-shell topologies, and a matching and optimization process that allocates the inventory to a target design and performs structural simulations. The methodology is demonstrated through one built design and several further design possibilities.

A generalized tooling strategy that doesn't require robotics and leverages mixed reality to enable the fabrication and assembly of irregular wooden elements into complex structures using only accessible woodworking tools. This strategy includes tolerance and joint manufacturing that improves versatility and accessibility. It is demonstrated in a gridshell case study built by three people.

Potential Impact

This research addresses the two challenges identified in building structures with wild wood: the design methodology and the fabrication process. These contributions could improve the effective consumption of unused timber resources and lay the foundation for exploring creative structural and architectural design opportunities. The presented method does not necessarily target adoption in a global timber industry

framework but rather addresses a postindustrial and decentralized practice of construction that has the potential to scale in a bottom-up fashion among a wider audience of nonexpert individuals and communities, as well as more traditional local actors. Furthermore, the research can be applied to various upcycling and structural design projects, enabling new construction practices with wild wood and any nonstandard elements. By facilitating upcycling in the design and construction of structures, this research can improve the material efficiency of buildings and reduce their associated embodied carbon.

Concluding Remarks

This paper presents a versatile and accessible method to compose and fabricate structures using wild wood. More generally, the research demonstrates how to leverage the idiosyncrasies of nonstandard elements to unlock new material sourcing for construction. The method facilitates upcycling approaches with computational matching, informing the design process and mixed reality fabrication. The computational tools and fabrication strategies presented can foster new design practices for the building industry to improve the efficiency of material consumption and participate in carbon mitigation.

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Data Statement

The data supporting this study's findings are available from the corresponding author, Tim Cousin, upon reasonable request.



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